Hundred days of coding c programming
Day-8

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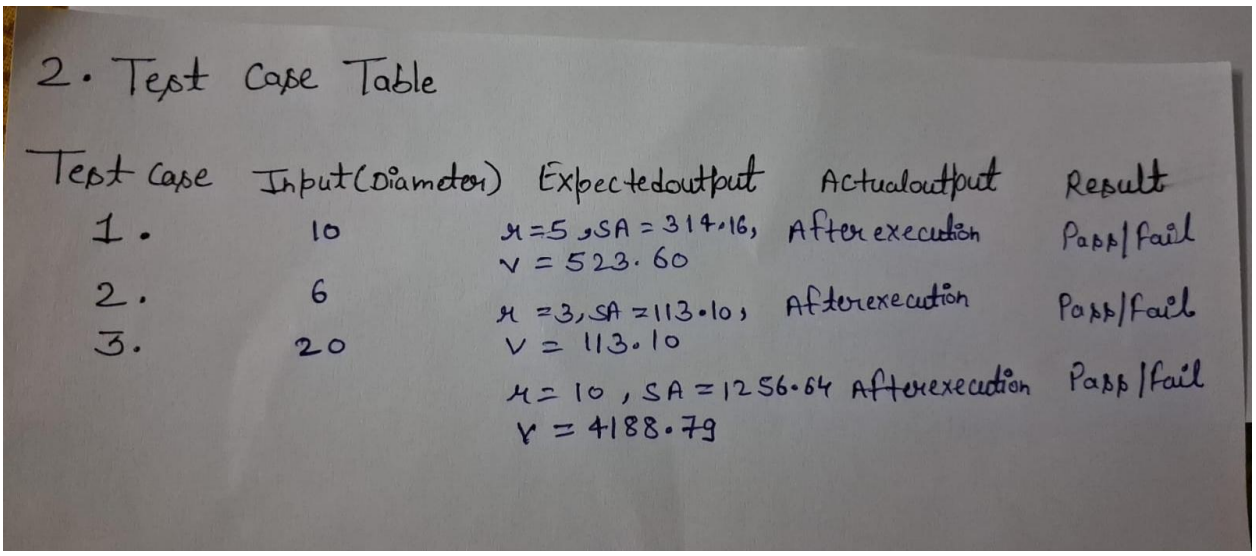
Question no. 1
Write a C program to find the radius, surface area, and volume of a sphere when its diameter is given.

1.**Understand the problem statement and identify the required input, output, and formula**.

Input:
Diameter of sphere = d Output:
Radius = r
Surface Area = SA
Volume = V Formula:
r = d / 2
SA = 4 × π × r × r
V = (4/3) × π × r × r × r Take
π = 3.14159.

2.**Prepare the standard test case table using the following format:**

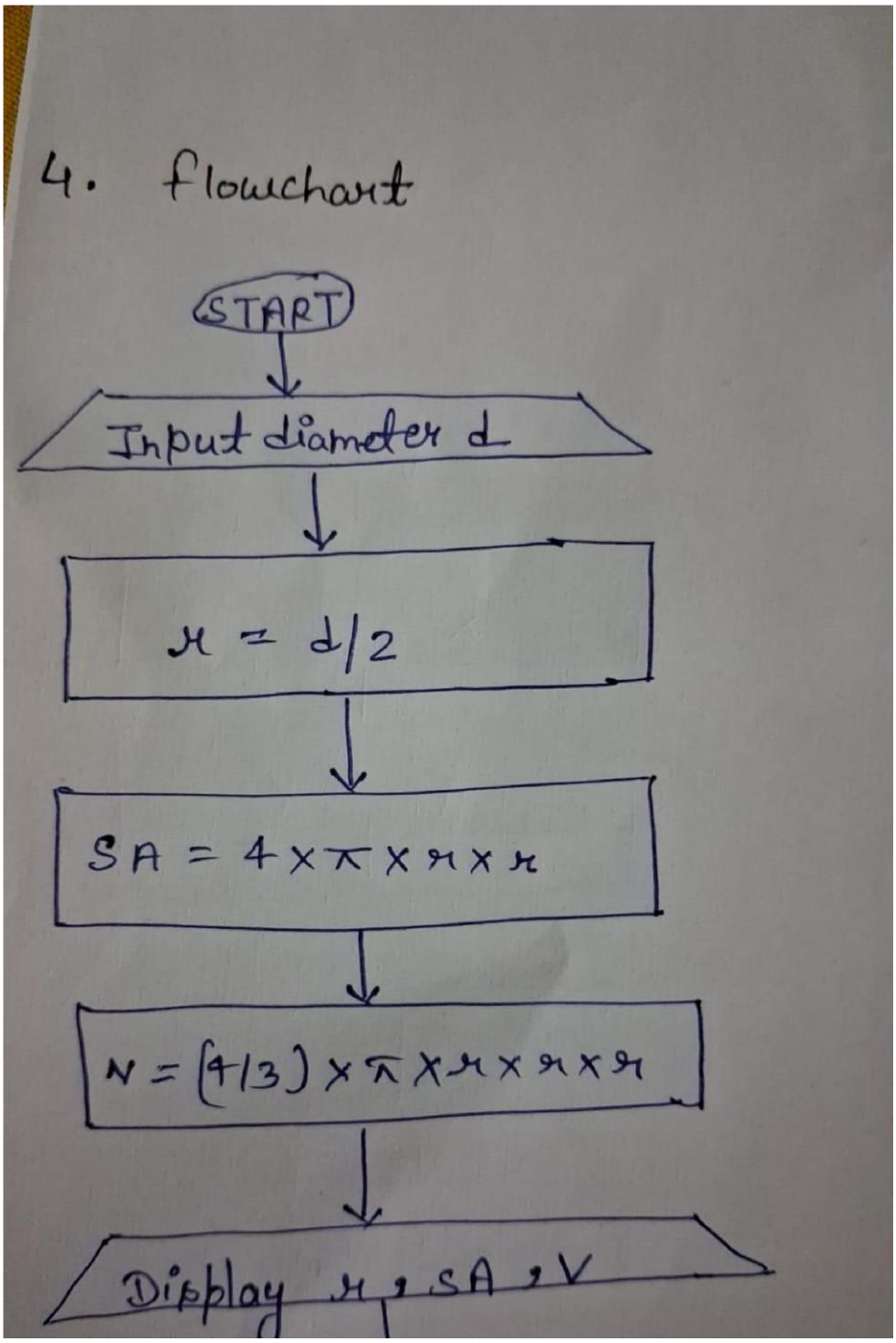


3.**Write (no snippets) and evaluate the algorithm against the test cases prepared in Step 2.**

. Algorithm to find radius, surface area and volume of sphere:

1.Start.
2.Read the diameter d.
3.Calculate radius using r = d / 2.
4.Calculate surface area using SA = 4 × π × r × r. 5.Calculate volume using V = (4/3) × π × r × r × r.
6.Display radius, surface area and volume.
Stop.

4.**Draw (no snippet) and evaluate the flowchart against the test cases prepared in Step 2.**



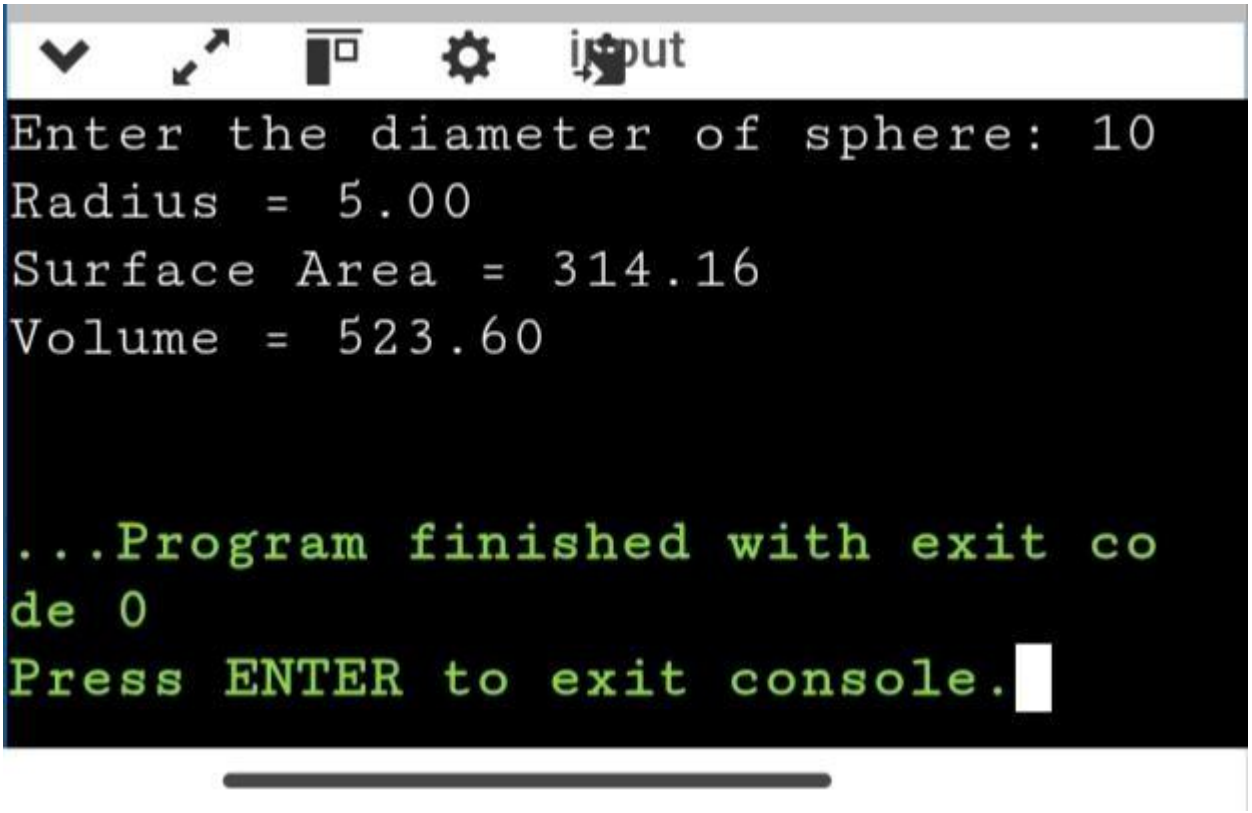
5.**Write (no snippets) the C program.**

ode

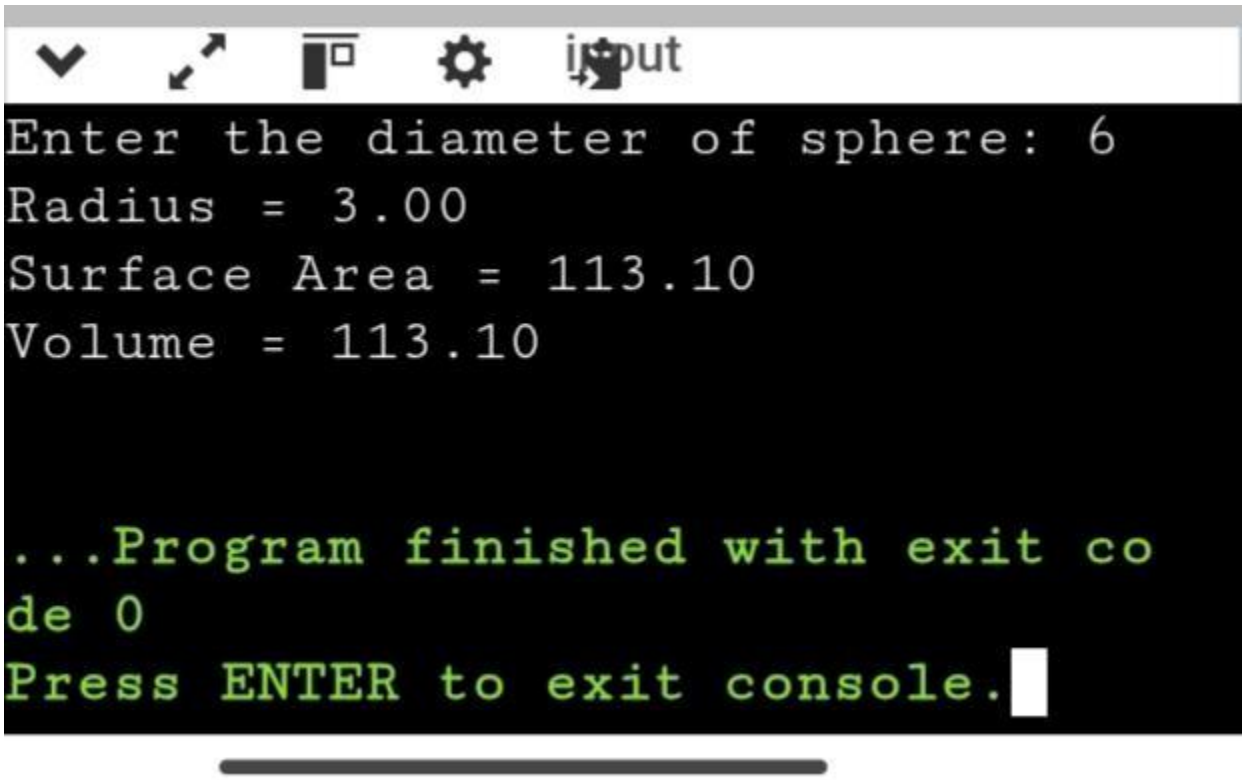
```c
#include <stdio.h>

int main()
{
    float d, r, sa, v;
    float pi = 3.14159;

    printf("Enter the diameter of sphere
    scanf("%f", &d);

    r = d / 2;
    sa = 4 * pi * r * r;
    v = (4.0 / 3.0) * pi * r * r * r;

    printf("Radius = %.2f\n", r);
    printf("Surface Area = %.2f\n", sa);
    printf("Volume = %.2f\n", v);

    return 0;
}
```

6.**Compile and execute the program, then evaluate the code against the test cases prepared in Step 2.**

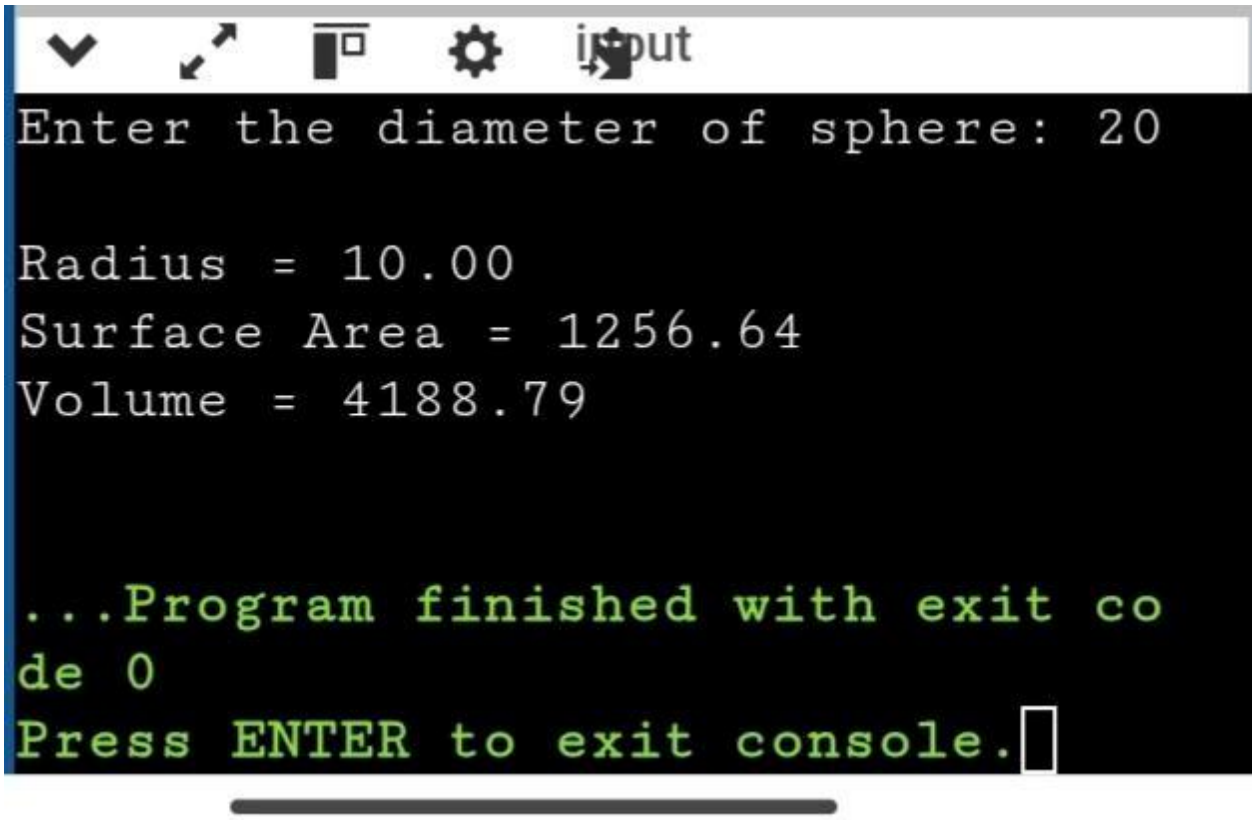
Case 1



Case 2


input
Enter the diameter of sphere: 6
Radius = 3.00
Surface Area = 113.10
Volume = 113.10
...Program finished with exit co
de 0
Press ENTER to exit console.

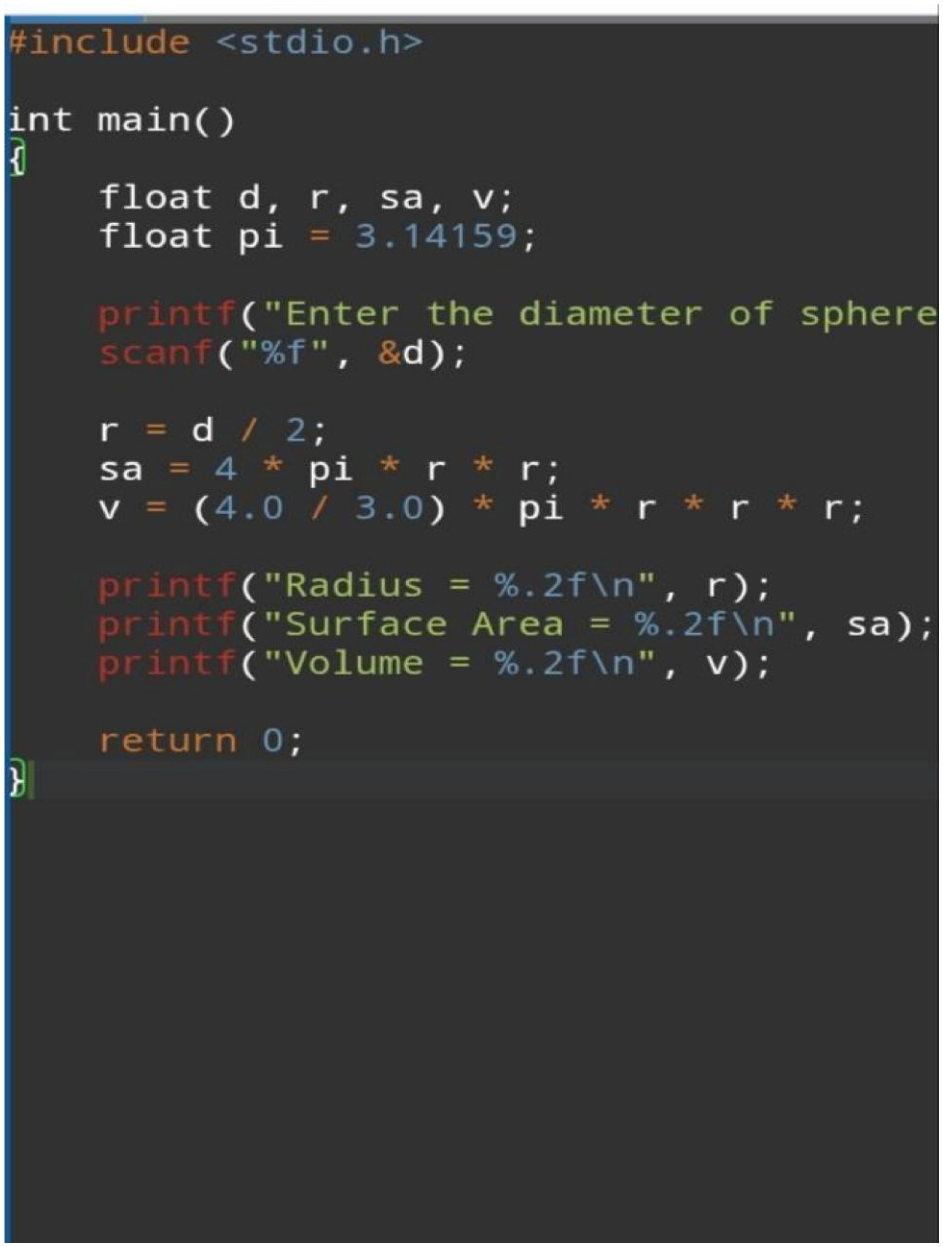
Case 3


input
Enter the diameter of sphere: 20
Radius = 10.00
Surface Area = 1256.64
Volume = 4188.79
...Program finished with exit co
de 0
Press ENTER to exit console.

7. Attach snippets/screenshots of compilation and execution.

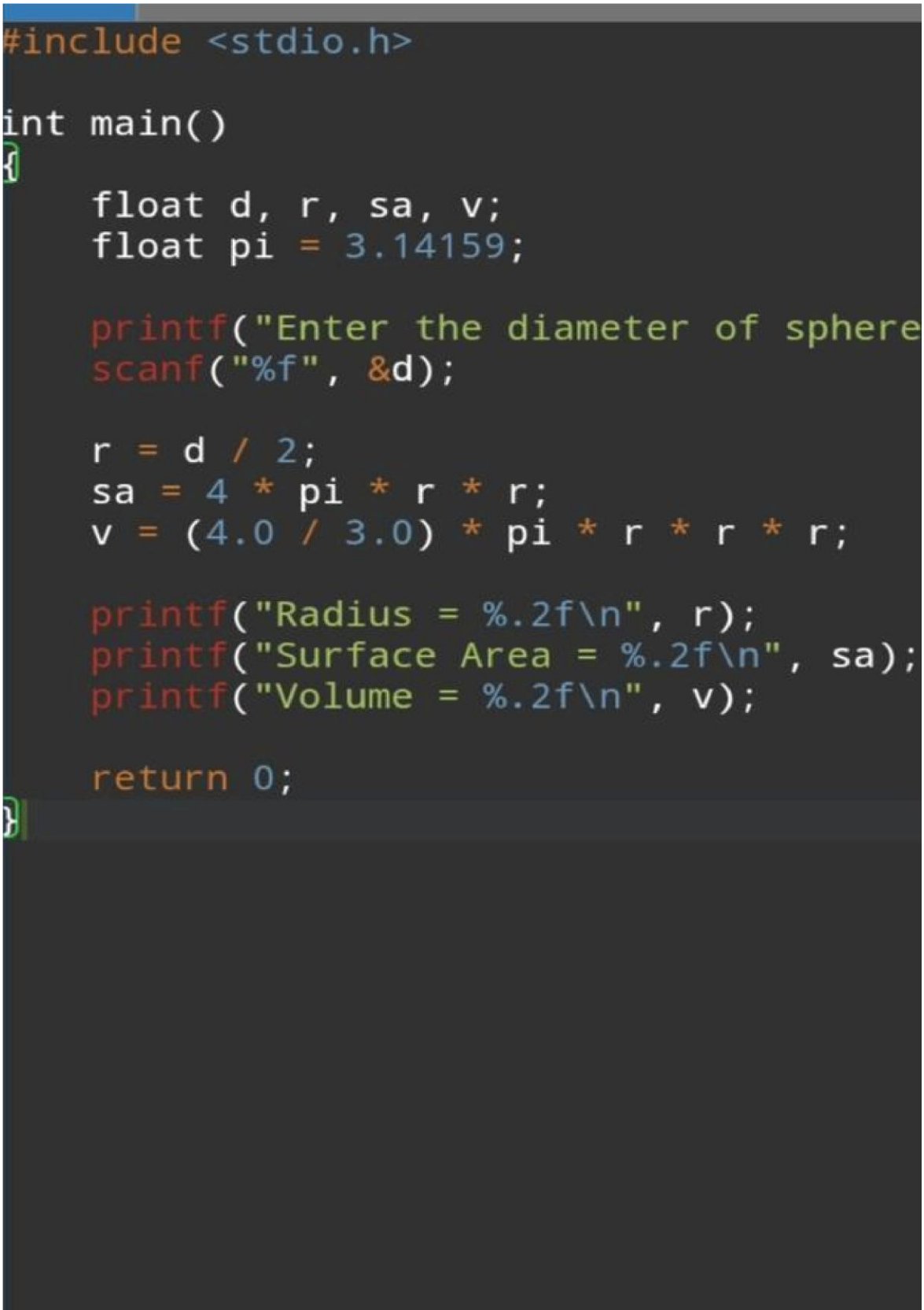
Compilation and Execution

CASE 1



```c
#include <stdio.h>

int main()
{
    float d, r, sa, v;
    float pi = 3.14159;

    printf("Enter the diameter of sphere
    scanf("%f", &d);

    r = d / 2;
    sa = 4 * pi * r * r;
    v = (4.0 / 3.0) * pi * r * r * r;

    printf("Radius = %.2f\n", r);
    printf("Surface Area = %.2f\n", sa);
    printf("Volume = %.2f\n", v);

    return 0;
}
```

CASE2



```c
#include <stdio.h>

int main()
{
    float d, r, sa, v;
    float pi = 3.14159;

    printf("Enter the diameter of sphere
    scanf("%f", &d);

    r = d / 2;
    sa = 4 * pi * r * r;
    v = (4.0 / 3.0) * pi * r * r * r;

    printf("Radius = %.2f\n", r);
    printf("Surface Area = %.2f\n", sa);
    printf("Volume = %.2f\n", v);

    return 0;
}
```

CASE 3

```c
#include <stdio.h>

int main()
{
    float d, r, sa, v;
    float pi = 3.14159;

    printf("Enter the diameter of sphere
    scanf("%f", &d);

    r = d / 2;
    sa = 4 * pi * r * r;
    v = (4.0 / 3.0) * pi * r * r * r;

    printf("Radius = %.2f\n", r);
    printf("Surface Area = %.2f\n", sa);
    printf("Volume = %.2f\n", v);

    return 0;
}
```

Conclusion

The C program successfully calculates the radius, surface area, and volume of a sphere when its diameter is given. The program was compiled and executed successfully, and the output matched the expected results for the test cases.

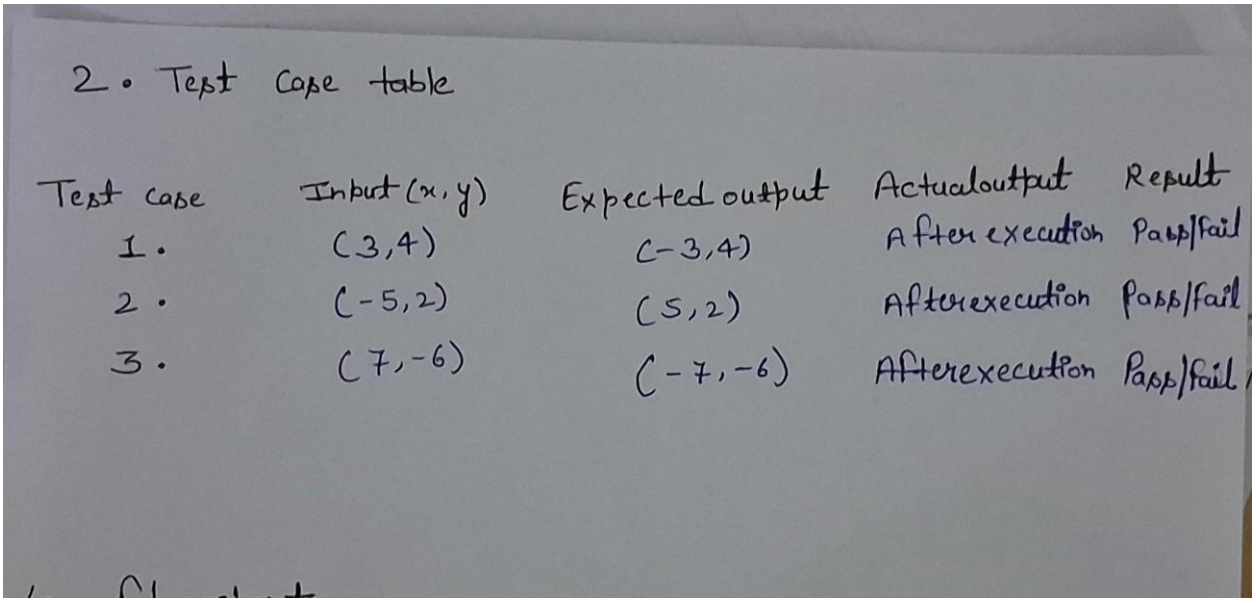
Question 2

Write a C program to find the image of a point after reflection in the y-axis.

1.Understand the problem statement and identify the required input, output, and formula.

Understand Problem
Input:
Coordinates of point (x, y) Output:
Coordinates of reflected point (x', y') Formula
for reflection in y-axis:
x' = -x y'
= y
Therefore,
Reflected point = (-x, y)

2.Prepare the standard test case table using the following format:



2. Test case table

Test case	Input (x, y)	Expected output	Actual output	Result
1.	(3,4)	(-3,4)	After execution	Pass/fail
2.	(-5,2)	(5,2)	After execution	Pass/fail
3.	(7,-6)	(-7,-6)	After execution	Pass/fail

3.Write (no snippets) and evaluate the algorithm against the test cases prepared in Step 2.

Algorithm

1.Start.

2.Read the coordinates x and y.

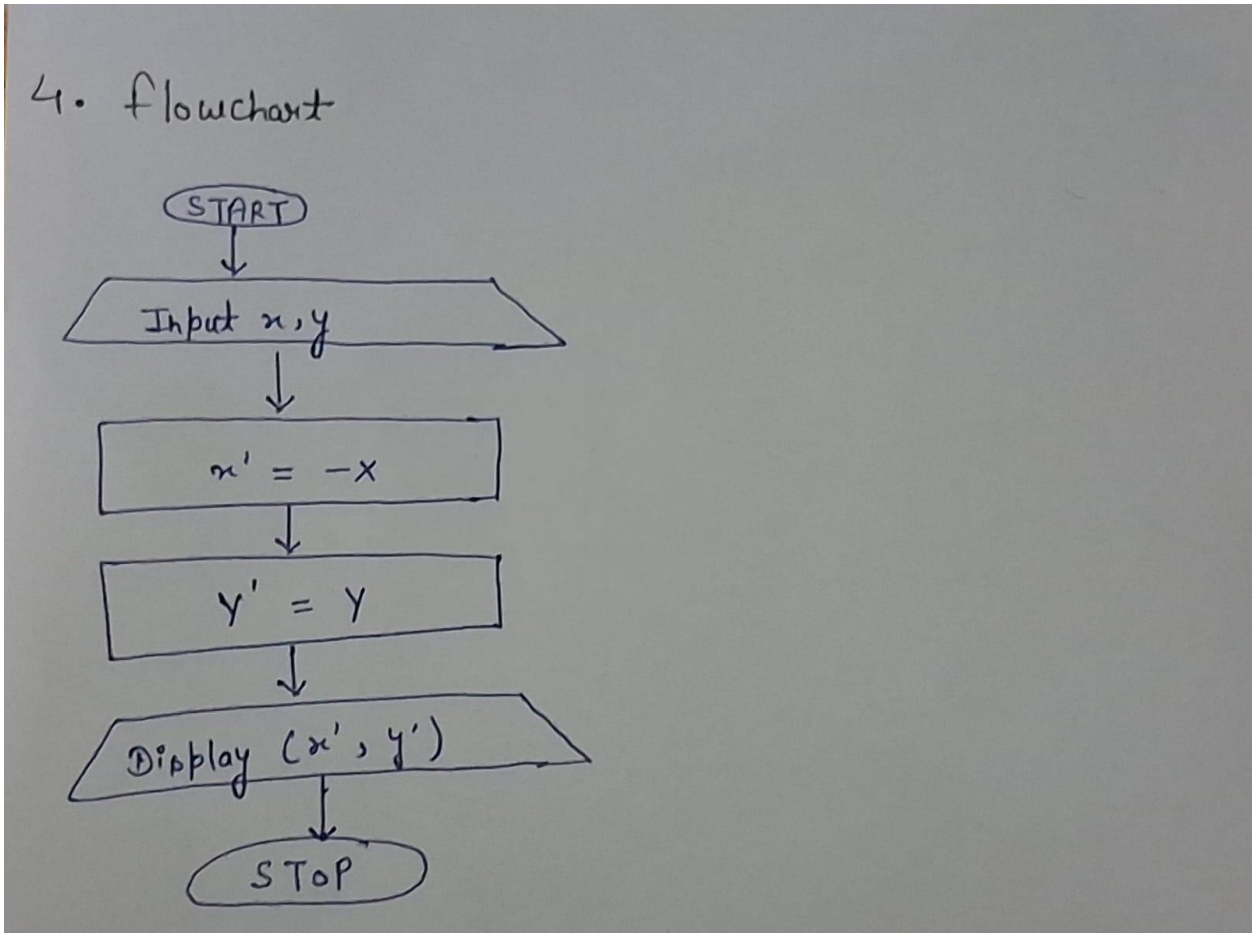
3.Calculate x' = -x.

4.Keep y' = y.

5.Display the reflected point (x', y').

6.Stop.

4.Draw (no snippet) and evaluate the flowchart against the test cases prepared in Step 2.



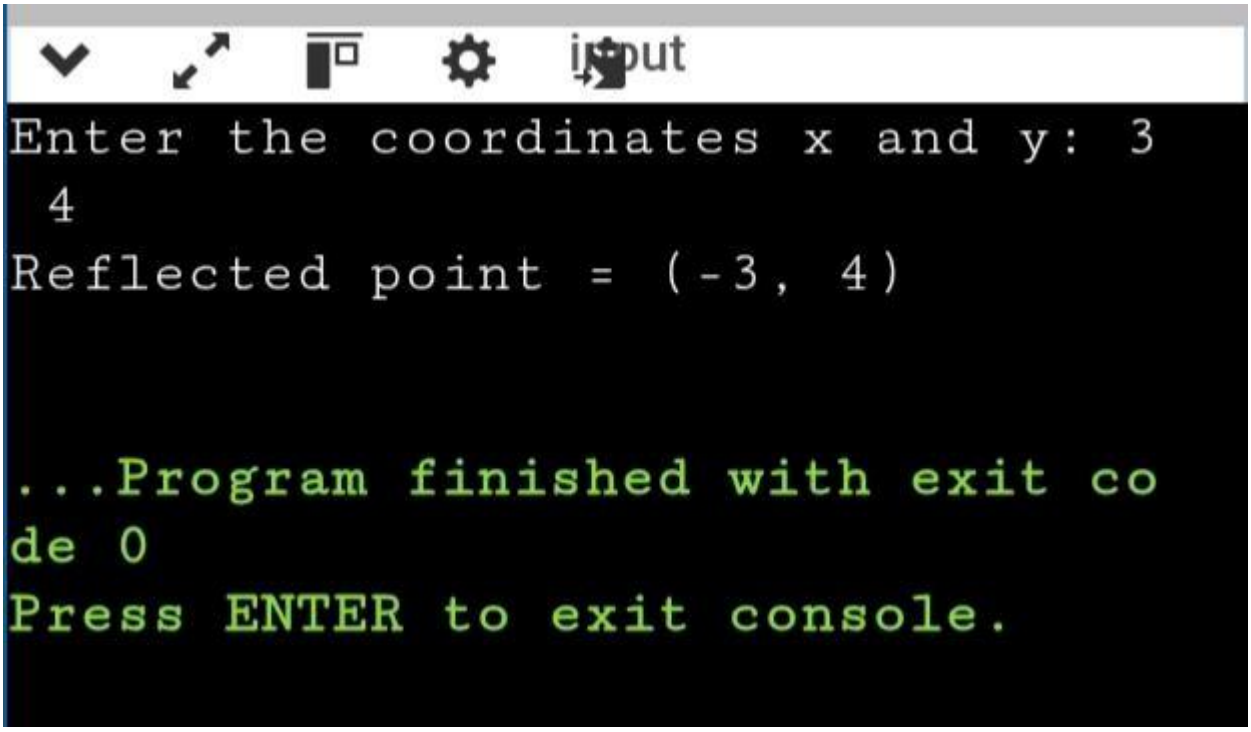
5.Write (no snippets) the C program.

main.c

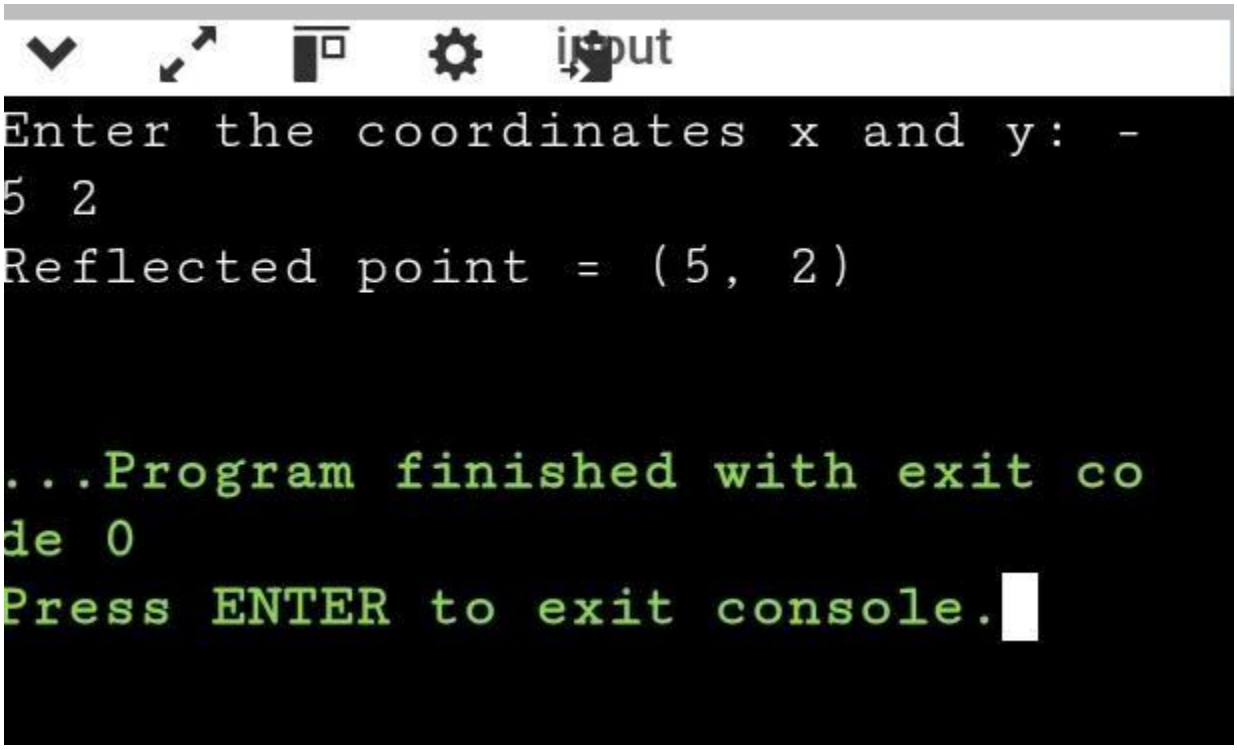
```c
#include <stdio.h>

int main()
{
    int x, y, x1, y1;

    printf("Enter the coordinates x and y: ");
    scanf("%d %d", &x, &y);

    x1 = -x;
    y1 = y;

    printf("Reflected point = (%d, %d)\n", x1, y1);

    return 0;
}
```

6.Compile and execute the program, then evaluate the code against the test cases prepared in Step 2.

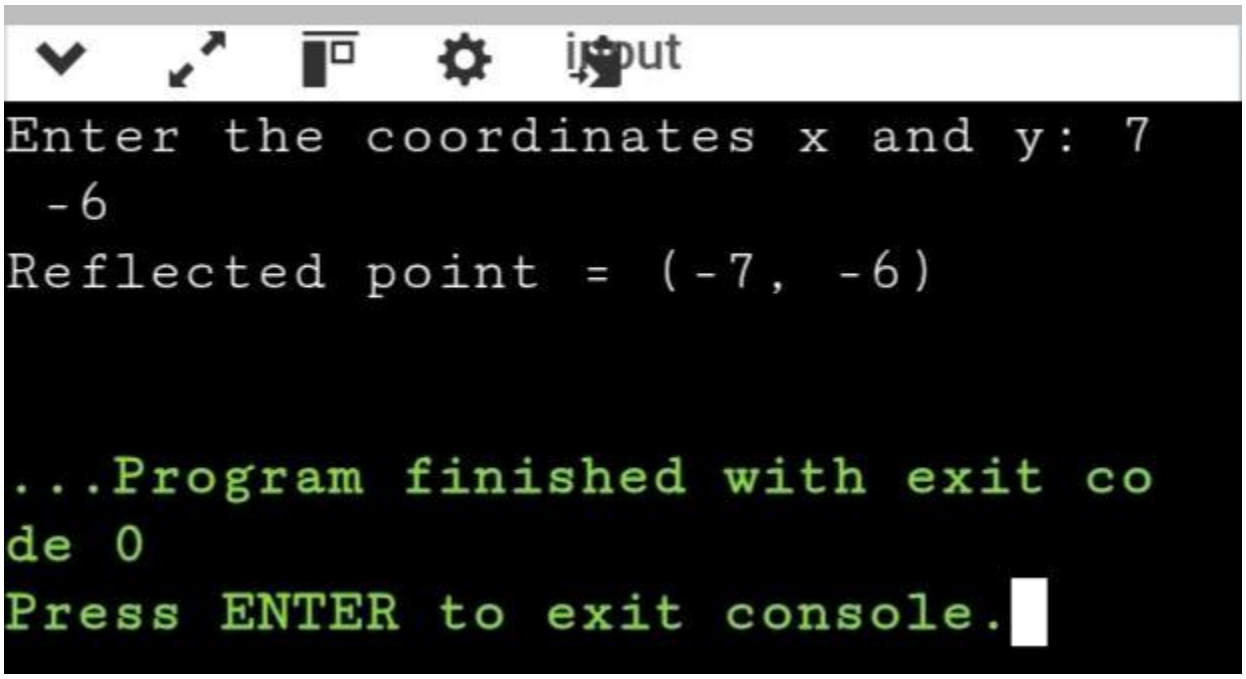
Case 1



Case 2



Case 3


input
Enter the coordinates x and y: 7
-6
Reflected point = (-7, -6)
...Program finished with exit co
de 0
Press ENTER to exit console.

7. **Attach Snippets/Screenshot of completion and execution.**

Compilation and Execution

CASE 1

main.c

```c
#include <stdio.h>

int main()
{
    int x, y, x1, y1;

    printf("Enter the coordinates x and y: ");
    scanf("%d %d", &x, &y);

    x1 = -x;
    y1 = y;

    printf("Reflected point = (%d, %d)\n", x1, y1);

    return 0;
}
```

```
Enter the coordinates x and y: 3,4
Reflected point = (-3, 1274077648)


...Program finished with exit code 0
Press ENTER to exit console.
```

CASE 2

main.c

```c
#include <stdio.h>

int main()
{
    int x, y, x1, y1;

    printf("Enter the coordinates x and y: ");
    scanf("%d %d", &x, &y);

    x1 = -x;
    y1 = y;

    printf("Reflected point = (%d, %d)\n", x1, y1);

    return 0;
}
```

```
Enter the coordinates x and y: -5,2
Reflected point = (5, -490303680)


...Program finished with exit code 0
Press ENTER to exit console.
```

CASE 3

main.c

```c
#include <stdio.h>

int main()
{
    int x, y, x1, y1;

    printf("Enter the coordinates x and y: ");
    scanf("%d %d", &x, &y);

    x1 = -x;
    y1 = y;

    printf("Reflected point = (%d, %d)\n", x1, y1);

    return 0;
}
```

```
Enter the coordinates x and y: 7,-6
Reflected point = (-7, 18420560)


...Program finished with exit code 0
Press ENTER to exit console.
```

Conclusion

The C program successfully finds the image of a point after reflection in the y-axis. The program correctly changes the sign of the x-coordinate while keeping the y-coordinate unchanged. The output matched the expected results for the test cases.